	Pimpri Chinchwad Education Trust's Pimpri Chinchwad College of Engineering (PCCoE) (An Autonomous Institute) Affiliated to Savitribai Phule Pune University(SPPU)	

Student Name- Yadnesh Patil
PRN -123B1D034

Self Learning Tasks

Title of the Case Study: Study of Postman – API Testing Tool for Software Development.

1. Introduction

In today's software engineering industry, modern applications depend on APIs (Application Programming Interfaces) for communication between frontend and backend systems. To ensure APIs work correctly, efficient testing tools are required. Manual API testing is time-consuming and error-prone, so automated tools are widely used.

This case study focuses on **Postman**, one of the most popular API testing and development tools. It helps developers design, test, debug, and document APIs efficiently.

2. Problem Statement

A company developing an Online Tourism Management System is facing the following challenges:

- Errors in booking and payment APIs
- Difficulty in testing backend services manually
- Slow communication between frontend and backend teams
- Delay in project delivery due to debugging issues

3. Objectives

- Study the working and structure of Selenium
- Study the working and use of AI in the field of software development
- Compare performance improvement achieved

4. System Architecture

4.1 Postman Architecture

Postman follows a client-server model.

Components:

- Postman Application (Desktop/Web)
- API Request Builder
- Server/API Endpoint
- Response Viewer
- Collection Runner

Workflow:

User Request → Postman → API Server → Response → Validation → Test Result

5. Working in SDLC

5.1 Requirement Phase

- Helps understand API requirements
- Defines request/response formats

5.2 Design Phase

- Creates API collections and endpoints
- Organizes APIs systematically

5.3 Development Phase

- Tests APIs during development
- Sends GET, POST, PUT, DELETE requests

5.4 Testing Phase

- Automates API testing using scripts
- Validates status codes and responses

5.5 Deployment & Maintenance

- Monitors APIs regularly
- Detects issues after deployment

6. Detailed Case Study Implementation
Application: Tourism Management System
Modules:

- User Authentication API
- Package Browsing API
- Booking API
- Payment API

6.1 Testing Using Postman

Test Cases:

1. User Login API
2. Package List Retrieval
3. Booking Confirmation
4. Payment Success Validation

Example Flow:

- Open Postman
- Select POST request
- Enter API URL
- Add JSON body
- Send request
- Verify response

7. Performance Analysis

Metric	Before Tools	After Tools
Testing Time	Manual, Slow	Fast
Bug Detection	Late	Early
Productivity	Moderate	High
API Validation	Difficult	Easy

- Advantages Easy to use interface
- Supports all HTTP methods
- Automated testing support
- API documentation generation
- Team collaboration features

8. Limitations

- Requires internet for cloud sync
- Limited for UI testing
- Complex automation needs scripting knowledge

9. Challenges Faced

- Learning API request structure
- Handling authentication tokens
- Managing dynamic data in requests

10.Risk Analysis

Risk	Impact	Mitigation
Wrong AI code	Medium	Verify endpoints
Test script failure	High	Use updated tokens
Incorrect Test Scripts	Medium	Debug Scripts

11.Future Enhancements

- AI-based API test generation
- Better automation integration
- Enhanced monitoring tools

12.Real-World Applications

- E-commerce platforms
- Banking systems
- Healthcare systems
- Travel and tourism platforms

13.Conclusion

Postman is a powerful API testing tool that improves software quality, reduces debugging time, and increases productivity. It plays an important role in modern software development by simplifying API creation, testing, and maintenance.

14. References

- Postman Official Documentation
- API Testing Tutorials
- Software Engineering Resources